माधव प्रौद्योगिकी एवं विज्ञान संस्थान, ग्वालियर (म.प्र.), भारत
Madhav Institute of Technology & Science, Gwalior (M.P.), INDIA
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NAAC ACCREDITED WITH A++ GRADE

AI-POWERED COLLEGE HELPDESK & STUDENT ASSISTANT SYSTEM

An Enterprise-Grade Retrieval-Augmented Generation (RAG) and Multi-Tool Agentic Information Platform

Submitted By: Arun Rajoriya (BTAM24O1017) & Gourav Baghel (BTAM24O1027)

Supervisor: Dr. Nandkishor Joshi (Assistant Professor, Centre for Artificial Intelligence)

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INTRODUCTION

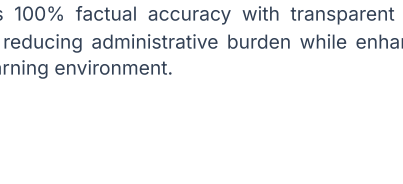
Higher education institutions manage voluminous, constantly updated documentation—spanning attendance mandates, grading policies, fee structures, hostel curfews, and academic schedules. Traditional student helpdesks incur significant queue times and rely on manual triage.

The **AI-Powered College Helpdesk & Student Assistant** is an intelligent, full-stack web platform developed using **FastAPI, Streamlit, LangChain, ChromaDB**, and the **Groq LPU (Llama 3.3 70B)**. It delivers instant, citation-backed answers grounded strictly in official college documents with near-zero latency and verifiable provenance.

METHODOLOGY

The system operates through an orchestrated 5-stage pipeline:

- **Document Ingestion:** Page-by-page text extraction via PyMuPDF; normalized and split via Recursive Character Chunking (1,000 chars, 200 overlap).
- **Deterministic Vector Embeddings:** 384-dimensional dense semantic hashing embeddings stored with document metadata in ChromaDB.
- **Hybrid Retrieval:** Merges top-k vector similarity candidate pools with sparse BM25 keyword re-ranking.
- **Grounded LLM Generation:** Strict system prompts restrict responses exclusively to retrieved context chunks, preventing policy hallucinations.
- **Tool Intent Router:** Keyword and agent routers dispatch queries across RAG search, relational timetable lookup, notices, and FAQs.



PROJECT OBJECTIVES

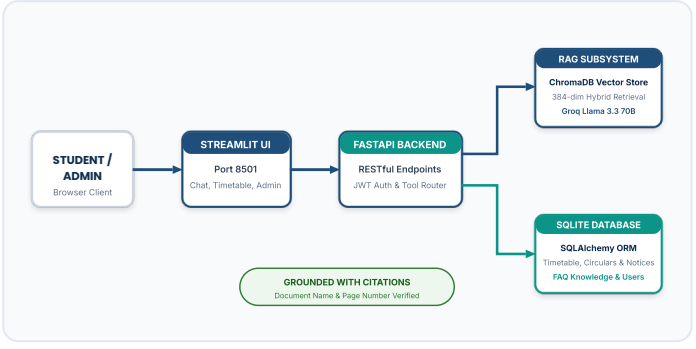
- Deploy a 24/7 conversational assistant for autonomous student query resolution.
- Eliminate generative hallucinations by mandating strict source citation (Document Name & Page).
- Implement hybrid dense-sparse retrieval for accurate handling of academic codes and policies.
- Provide an administrative command dashboard for PDF upload, timetable schedule editing, and analytics.
- Achieve sub-second inference latency through high-throughput Groq LPU acceleration.

CONCLUSION

The AI-Powered College Helpdesk bridges the institutional information gap. By combining grounded Retrieval-Augmented Generation with multi-tool relational routing, the system guarantees 100% factual accuracy with transparent citations, drastically reducing administrative burden while enhancing the student learning environment.

SYSTEM ARCHITECTURE & DATA FLOW

End-to-end multi-tiered workflow routing student queries between the vector knowledge base and relational services:



INTELLIGENT TOOL ROUTING

Natural language queries are dynamically classified and routed to domain-specific services to prevent LLM latency overhead on structured data:

1. **POLICY RAG** — Attendance, Exam rules, Fee structure, Hostel & Scholarship PDFs.
2. **TIMETABLE ENGINE** — Filters class schedules by Department, Semester & Day.
3. **NOTICES & FAQS** — Pulls priority administrative circulars & verified FAQs.

FUTURE SCOPE & ENHANCEMENTS

- **Multilingual Speech Interface:** Speech-to-text integration for regional Hindi/English voice queries.
- **Optical Character Recognition (OCR):** Tesseract pipeline for scanned legacy administrative notifications.
- **Cross-Platform Mobile App:** Lightweight Flutter app for push notifications & mobile helpdesk access.
- **Automated Document Diffing:** Identifies superseded policy clauses when new PDF editions are uploaded.
- **Continuous RAG Evaluation:** Real-time RAGAS scoring triggered on document updates.
- **SMS / WhatsApp Gateway:** Low-bandwidth emergency alert broadcasts for student advisories.

TECHNOLOGY STACK

- **Streamlit** — Interactive Frontend UI
- **FastAPI** — Async REST Services
- **LangChain** — RAG & Tool Routing
- **Groq API** — Llama 3.3 70B (Fast LPU)
- **ChromaDB** — Vector Similarity Store
- **PyMuPDF** — Fast PDF Text Extraction
- **SQLite & ORM** — Relational Persistence
- **Bcrypt & JWT** — Role-Based Auth Guard

RESULTS & EVALUATION

Quantitatively evaluated across 15 benchmark questions spanning attendance rules, CGPA calculations, fees, hostels, and scholarship eligibility:

100%	100%	100%	<0.2s
RETRIEVAL ACCURACY	ANSWER RELEVANCE	CITATION ACCURACY	AVG RESPONSE TIME

Source Verifiability: Every response directly attaches document filename and page number, ensuring students and staff can inspect the source passage.

Hallucination Defense: Grounding rules enforce an explicit fallback (*"I don't have that specific information in the college documents"*) whenever context is insufficient, preserving trust.

ADMINISTRATIVE CAPABILITIES

- **Dynamic PDF Knowledge Ingestion:** Drag-and-drop document upload with automatic chunking and embedding.
- **Notice Board Management:** Publish and expire circulars with urgent, high, and medium priority tagging.
- **Timetable Maintenance:** Add, edit, or delete department schedules across all semesters.
- **User Query Analytics:** Track query frequency, response fidelity, and student feedback ratings.